

Tree Deletion

Deleting Items From Trees

```
def delete_item(self, item) -> bool:
    if self.is_empty():
        return False # item isn't in the tree
    elif self._root == item:
        self._delete_root()
        return True # item was deleted
    else:
        ... # recurse somehow
```

- This template assumes that `self._subtrees` contains no empty subtrees
 - Can be shown as a Representation Invariant
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Reminders and Tips

- Revisit readings, especially notes on mutating trees
- Run the code and add tests to convince yourself
- Partially trace the recursive step
- Analyze the code
 - Ex. What happens if `_subtrees` contains empty trees
- Ask yourself questions, try out things, test, trace, analyze, etc.